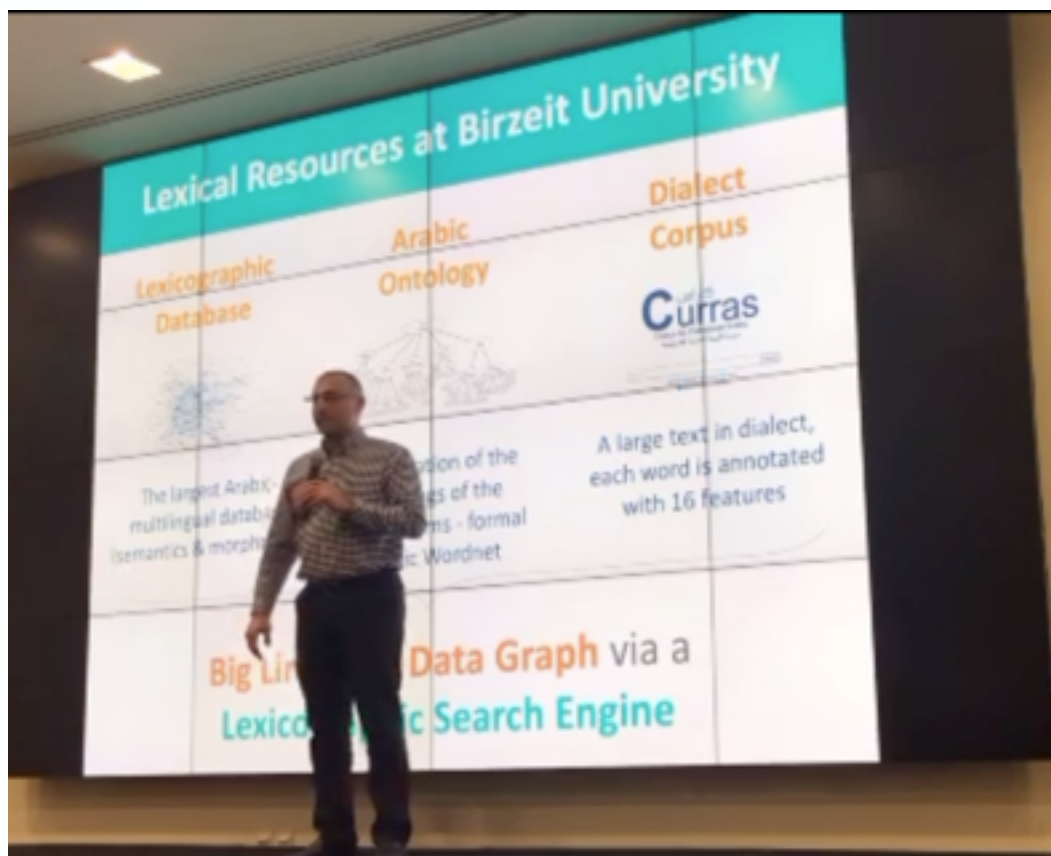


Towards building a **Big Linguistic Data Graph**

Mustafa Jarrar
Birzeit University
Palestine

Watch the lecture here



<https://twitter.com/ArabicSpeech/status/1122843057756475392?s=20>

Why Lexical Resources!

- ❖ The importance of lexical resources (dictionaries, thesauri, wordnets, linguistic ontologies) is increasing in many application areas, such as:
 - NLP tasks and applications
 - Information search and retrieval
 - Multilingual big data
 - Multilingual semantic web
 - Data integration
 - among many others.
- ❖ Lack of Arabic Lexical resources for human use!
- ❖ Lack of Arabic Lexical resources for NLP!

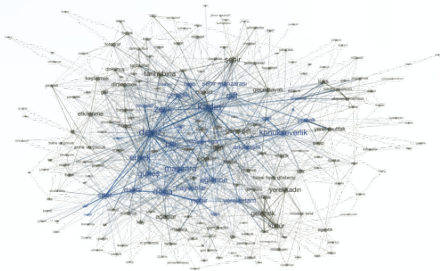
Digitize, Collect, Build, then clean and link



- **Make available online for people**
- **Make available through APIs for NLP applications**

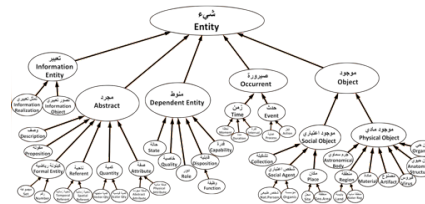
Lexical Resources at Birzeit University

Lexicographic Database



The largest Arabic-multilingual database (semantics & morphology)

Arabic Ontology



Classification of the meanings of the Arabic Terms - formal Arabic Wordnet

Dialect Corpus



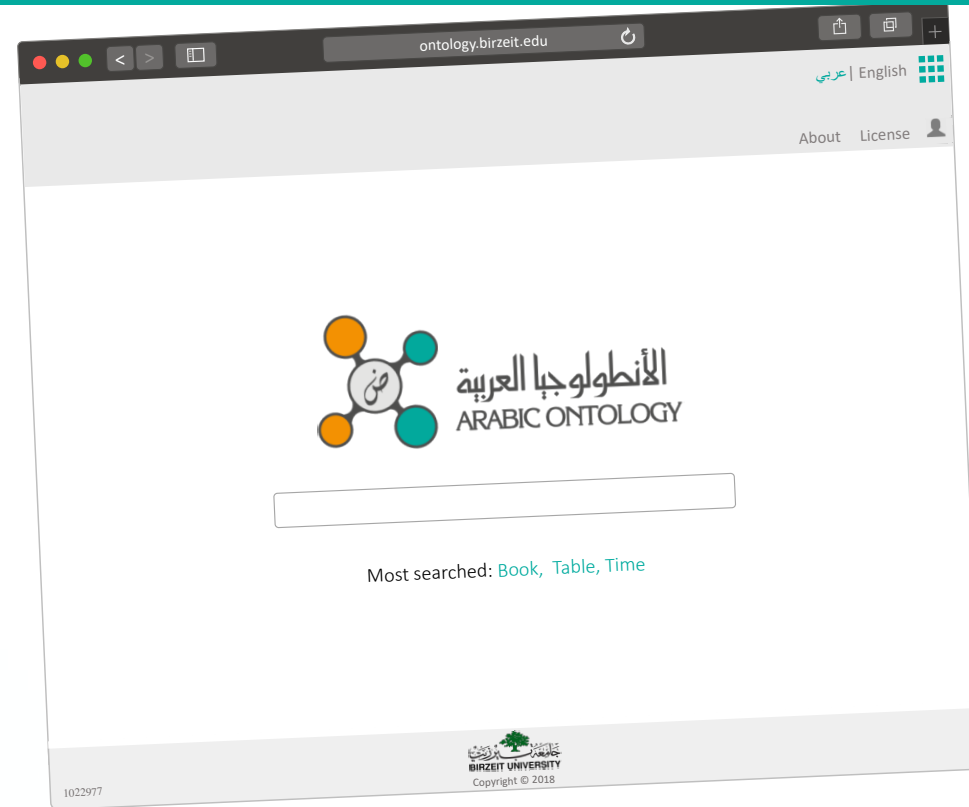
A large text in dialect, each word is annotated with 16 features

Big Linguistic Data Graph via a
Lexicographic Search Engine

Lexicographic Database

Lexicographic Search Engine

- **Free access to people:** students, translators, researchers, Arabic learners ...
- **API accessible** for NLP applications.



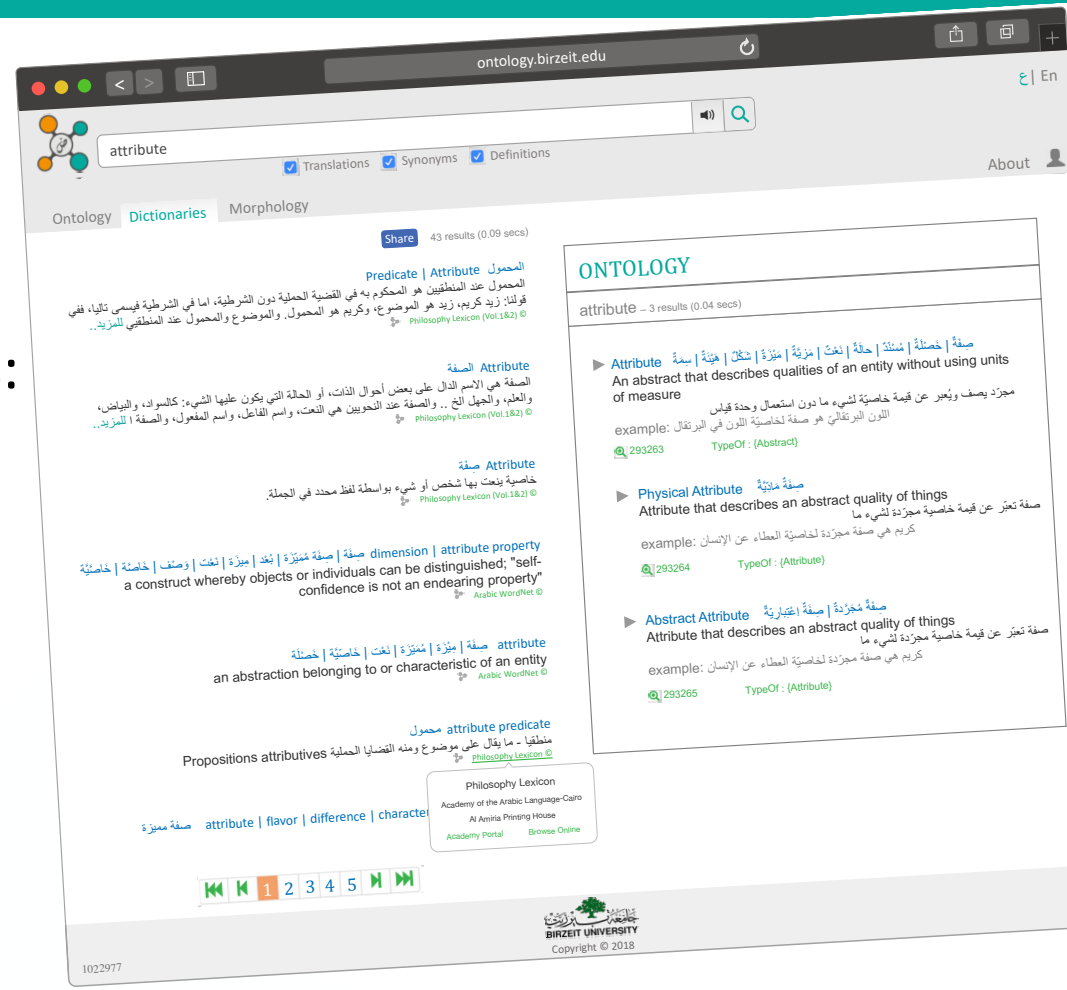
<https://ontology.birzeit.edu>

Based on:

Mustafa Jarrar, Hamzeh Amayreh: **An Arabic Multilingual Database with a Lexicographic Search Engine**. The 24th International Conference on Applications of Natural Language to Information Systems (NLDB 2019), Springer. Manchester, UK. 2019

Lexicographic Search Engine

- The largest lexicographic Arabic database
- Contains most lexicon types: glossaries, thesauri, bi/trilingual dictionaries, morph datasets, **Arabic Ontology**, and more.
- Covers most domains: science, technology, law, business, art, philosophy, ...



<https://ontology.birzeit.edu>

Lexicographic Search Engine

- **Search 150 lexicons** for definitions, synonyms and specialized translations.
- **Accurate!** compared with machine translation.
- **The first of its kind!** e.g., there are no similar search engines for English lexicons



DEMO

<https://ontology.birzeit.edu>

Lexicographic Search Engine

Conformance with W3C Standards:

- ✓ **W3C's Best Practices for Publishing Linked Data**
including the Cool URIs, simplicity, stability, and linking
- ✓ **W3C's RDF Lemon Model**

التسوية levelling | grading

تحريك التربة أثناء إعداد الأرض للري للوصول إلى سطح مستو أو سطح ذي انحدار منتظم.

 Hydrology Lexicon ©

```
...
@prefix aot: <http://ontology.birzeit.edu/term/>.
@prefix aoc: <http://ontology.birzeit.edu/lexicalconcept/>.
@prefix aor: <http://ontology.birzeit.edu/lexicon/>.

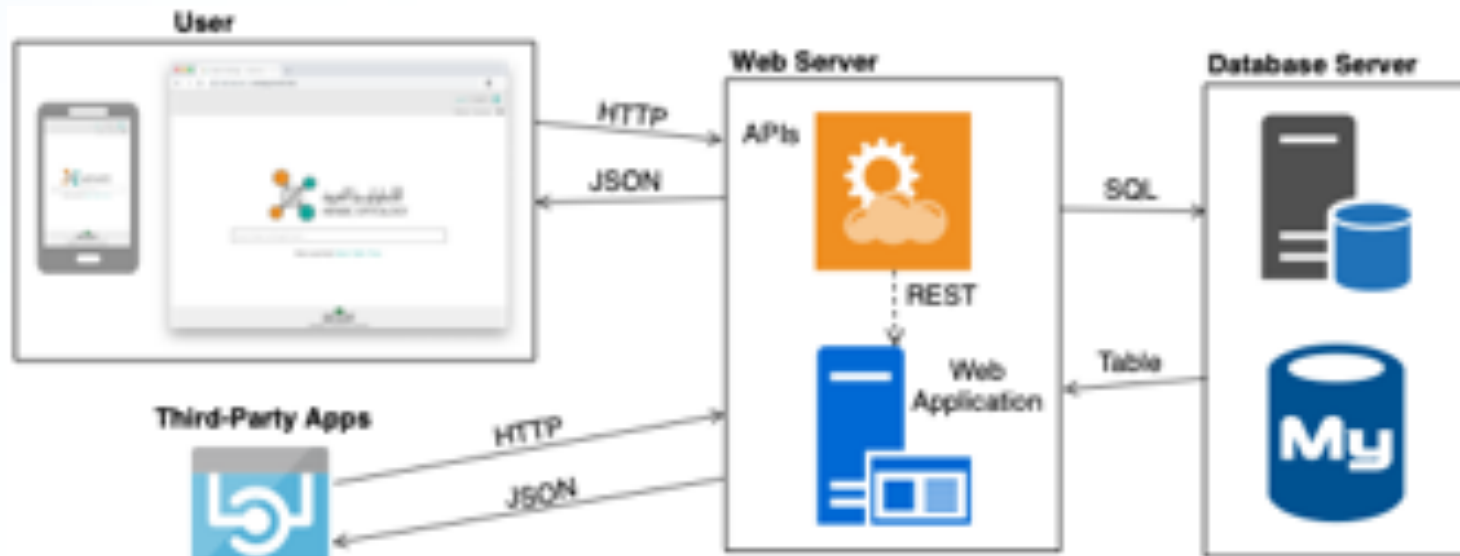
<aoc:1623> a ontolex:LexicalConcept;
ontolex:isEvokedBy <aot:Lex-grading>;
ontolex:isEvokedBy <aot:Lex-levelling>;
ontolex:isEvokedBy <aot:Lex-تسوية>;
skos:definition "تحريك التربة أثناء إعداد الأرض للري للوصول إلى سطح مستو أو سطح..."@ar;
skos:inScheme <aor:Hydrology_Lexicon_1>.

<aot:lex-grading> a ontolex:LexicalEntry, ontolex:Word;
ontolex:canonicalForm [ontolex:writtenRep "grading"@en];
skos:inScheme <aor:Hydrology_Lexicon_1>.
<aot:lex-levelling> a ontolex:LexicalEntry, ontolex:Word;
ontolex:canonicalForm [ontolex:writtenRep "levelling"@en];
skos:inScheme <aor:Hydrology_Lexicon_1>.
<aot:lex-تسوية> a ontolex:LexicalEntry, ontolex:Word;
ontolex:canonicalForm [ontolex:writtenRep "تسوية"@ar];
skos:inScheme <aor:Hydrology_Lexicon_1>.
```

Based On:

Mustafa Jarrar, Hamzeh Amayreh, John McCarae: **Progress on Representing Arabic Lexicons in Lemon**. The 2nd Conference on Language, Data and Knowledge (LDK 2019), Germany. 2019.

Lexicographic Search Engine



Search Engine Architecture

Lexicographic Search Engine

Some Statistics

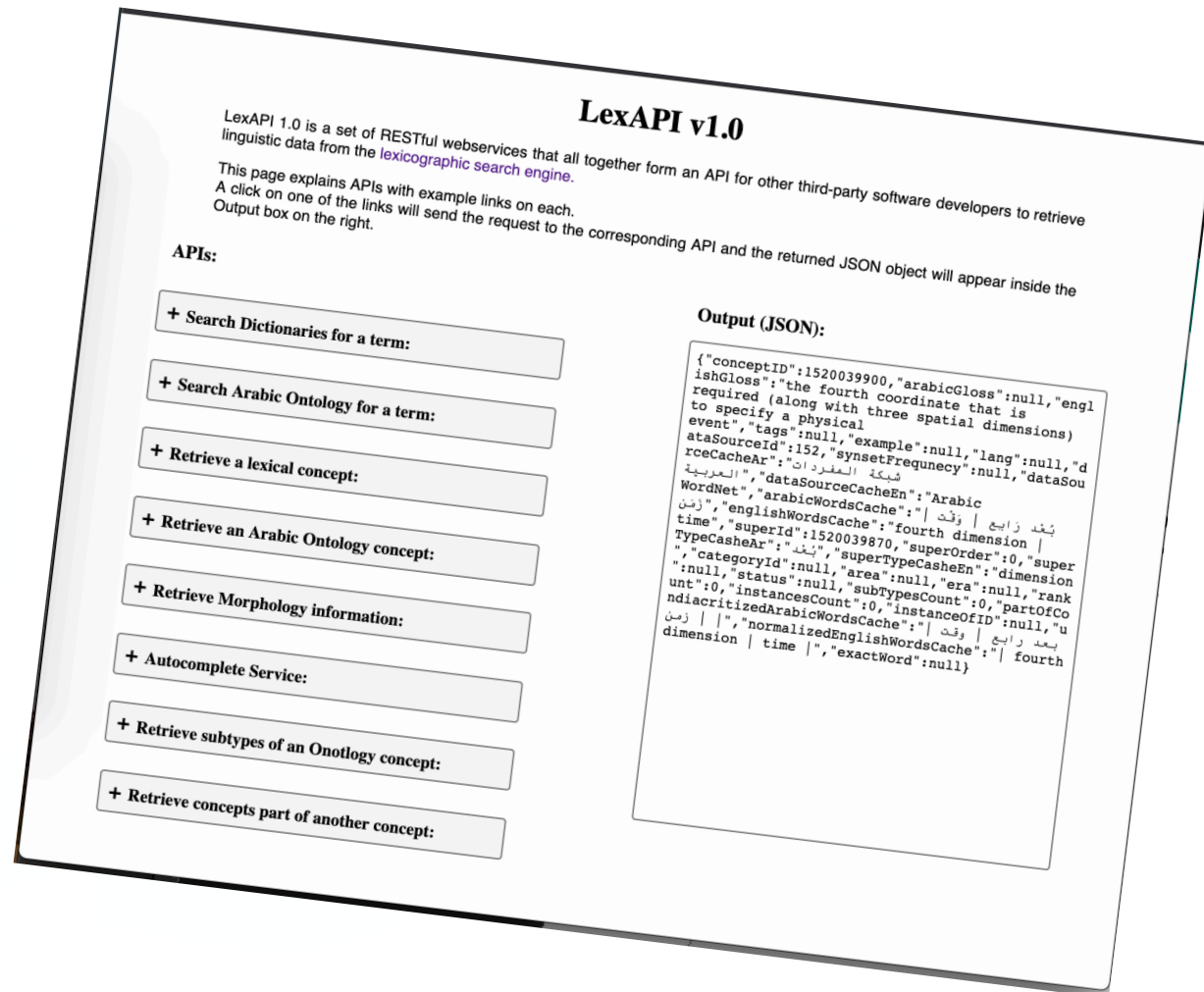
Category	Lexical Concepts	Lexical entries	Synsets	Translations pairs	Glosses	Semantic relations
Total (Millions)	1.1 M	2.4 M	1.8 M	1.5 M	0.7 M	0.5 M
Sub Counts		1,100K Arabic 1,100K English 200K French 3K Others 1,300K Single-word 1,000K Multi-word	800K Arabic 800K English 200K French 50K Others	1,000K English-Arabic 300K English-French 200K French-Arabic	400K Arabic 300K English 1K Others	170K Sub-super links 29K Part-of links 260K Has-Domain links 30K Other links

Lexicographic Search Engine

API Access

RESTful web services

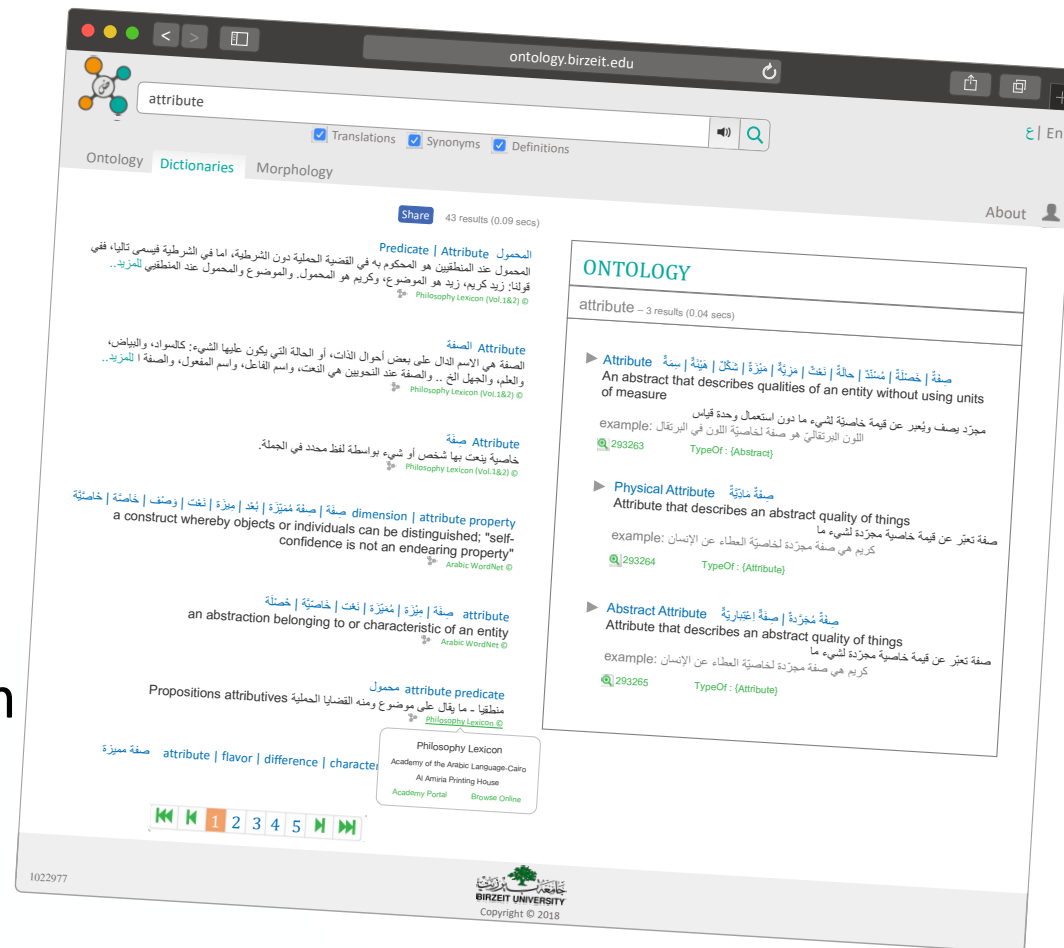
Ask us for an API Key!



Lexicographic Search Engine

Done over 9 years:

- Obtain permission from each dictionary owner.
- Manually type, then structure and store into one database.
- Didn't receive any fund from any party!
- Non-profit research



Arabic Ontology

Arabic Ontology

- Classification of the meanings of the Arabic terms, specified in D. Logic
- Can be used as a formal Arabic Wordnet -with ontologically-clean content.
- Linked with WordNet
- Benchmarked to scientific advances rather than to speakers' naïve beliefs as wordnets do.

ع | En

Translations Synonyms Definitions

About

Ontology Dictionaries Morphology

شيء | كَيُونَة | كَان Entity
Whatever existed or will exist, and can be realized or imagined
أَيَّمَا وُجِدَ أَوْ سَيُوجَدُ وَنَسْتَطِيعُ إِدْرَاكَهُ أَوْ تَخَيُّلَهُ
example: كل شيء على ما يرام 293198

مَوْجُود | كَان | قَانِم | حَقِيقِي | واقعي | شيء | ذات | قِيَوْم Object
An entity that is wholly and independently present in time, and is realized either for its concrete or social existence
شيء له ذات مستقلة بنفسه، وحاضر كلياً في الزمن، ويُدرَك بذاته قياساً أو لذاته اعتباراً
example: يختلف إدراكنا لأي موجود لاختلاف ما يميّز أنواعه من الصفات الجوهرية 293200 TypeOf: {Entity}

صَيَّرُوْرَة | حَدَث | حَادِث | وَقَع | أَمْر Occurrent
An entity realized by the time of its happening
شيء يدرك ذاته بزمان حدوثه
example: لا يمكن فهم أي حدث بشكل منفصل عن الإطار الزمني له 293202 TypeOf: {Entity}

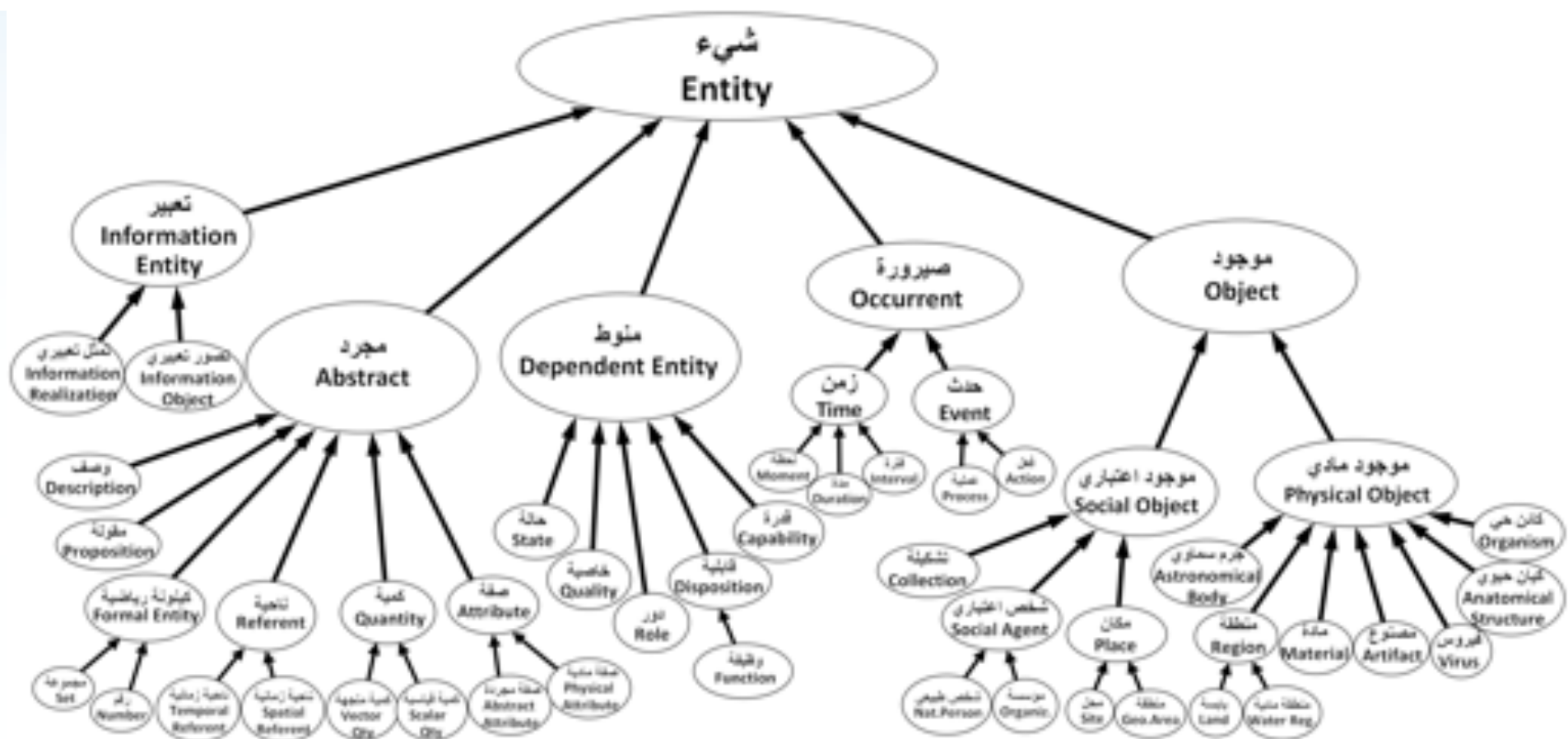
مَنْوُوط | مَعْتَمِد | مُتَعَلِّق | مَشْرُوط Dependent Entity
An entity realized by the time of its happening
شيء يعتمد وجوده على وجود أشياء أخرى
example: طول المبنى منوط بوجود المبنى وإلا فلا طول له 293201 TypeOf: {Entity}

مُجَرَّد | تَجْرِيْدِي | غَيْر مَادِّي | نَظَرِي Abstract
An entity exists only in mind, cannot be measured or socially realized, and does not have a location
1022977

BIRZEIT UNIVERSITY
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<https://ontology.birzeit.edu/concept/293198>

Arabic Ontology



Based on:

Mustafa Jarrar: **The Arabic Ontology - An Arabic Wordnet with Ontologically Clean Content.** Applied Ontology Journal, IOS Press. (Forthcoming).

Dialect Corpus

Dialect Corpus

<http://portal.sina.birzeit.edu/curras>

- Collect a corpus written in Palestinian dialect.
- Describe and annotate each word with 16 tags.
- Stats:
60k tokens,
16K unique tokens



Based on:

Mustafa Jarrar, Nizar Habash, Faeq Alrimawi, Diyam Akra, Nasser Zalmout: Curras: **An Annotated Corpus for the Palestinian Arabic Dialect**. Journal Language Resources and Evaluation. Volume(51), Issue(3). Springer. 2017

<http://portal.sina.birzeit.edu/curras>

[illegible]

Dialect Corpus

Middle size!

Size

Resource	Tokens (%)	Documents
Facebook	4852 (8.6)	35 threads
Twitter	4694 (8.3)	38 threads
Blogs	11,245 (19.8)	37 threads
Forums	1027 (1.8)	33 threads
Palestinian stories	3149 (5.6)	6 stories
Palestinian terms	1468 (2.6)	1 doc
TV Shows: وطن ع وثر <i>Watan Aa Watar</i>	30,265 (53.4)	41 episodes
Total	56,700	190

Accuracy of annotators

Category	A1 accuracy	A2 accuracy
Complex-prefix	97.4	97.6
Stem	86.3	93.1
Complex-suffix	92.5	96.0
Surface (CODA)	90.8	97.8
Lemma	93.4	93.1
Lemma _{MSA}	87.5	87.2
Gloss	74.6	80.9
POS	94.1	95.7
Person	97.8	98.8
Aspect	99.3	99.1
Gender	86.2	96.9
Number	94.8	96.8

Accurate



Connecting
lexical resources

W3C Lemon RDF Model Standard

Lexicon Model for Ontologies: Community Report, 10 May 2016



Final Community Group Report 10 May 2016

Editors:

[Philipp Cimiano](#) ([Cognitive Interaction Technology Excellence Center, Bielefeld University](#))

[John P. McCrae](#) ([Insight Centre for Data Analytics, National University of Ireland, Galway](#))

[Paul Buitelaar](#) ([Insight Centre for Data Analytics, National University of Ireland, Galway](#))

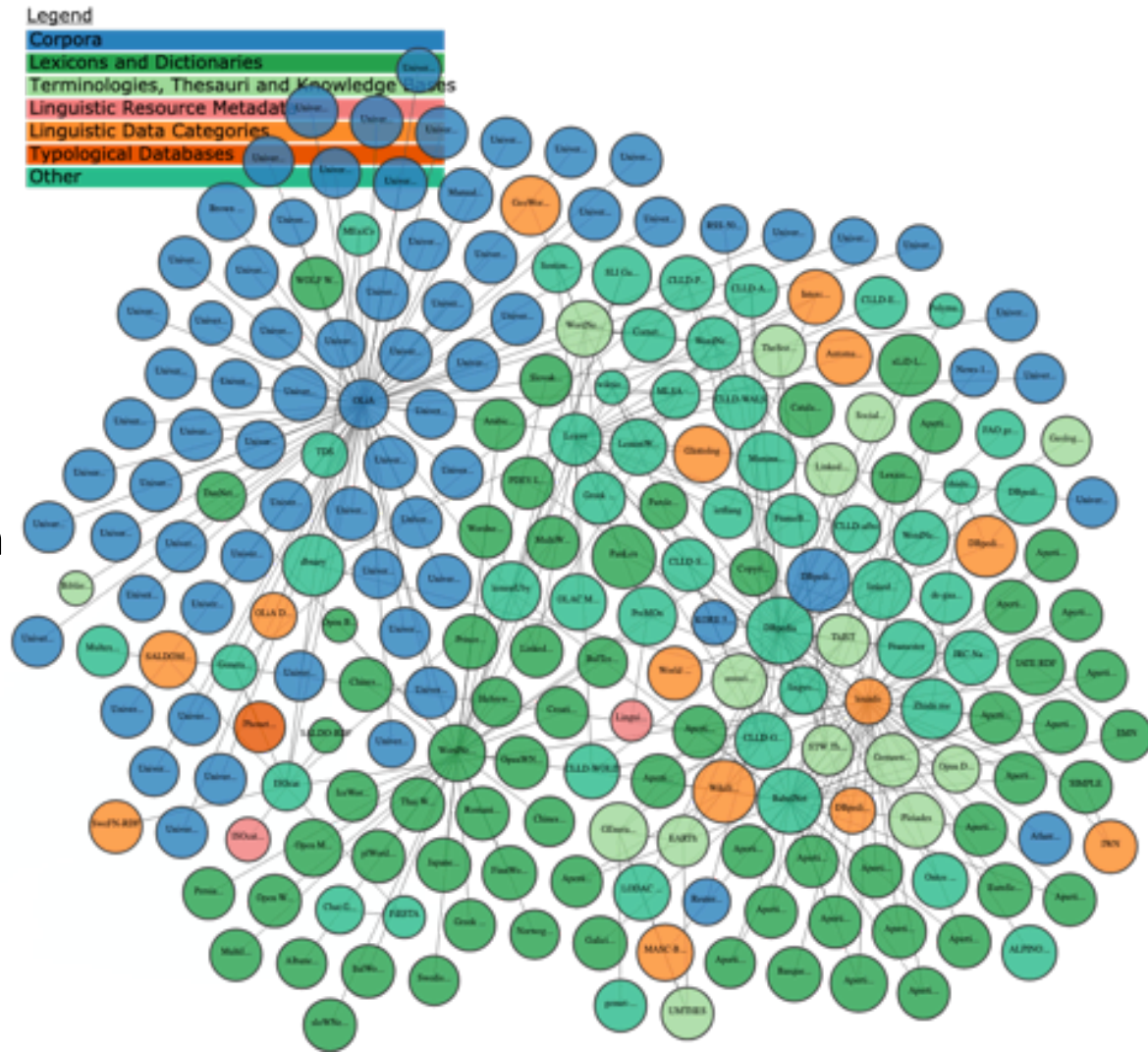
[Copyright](#) © 2016 the Contributors to the Lexicon Model for Ontologies: Community Report, 10 May 2016 Specification, published by the [Ontology-Lexicon Community Group](#) under the [W3C Community Final Specification Agreement \(FSA\)](#). A human-readable [summary](#) is available.

Abstract

This document describes the lexicon model for ontologies (*lemon*) as a main outcome of the work of the Ontology Lexicon (Ontolex) community group.

The Linguistic Linked Open Data Cloud

- A collaborative effort to develop a Linked Open Data (sub-)cloud of linguistic resources.
- Represent (lexical entries, concepts, synsets, and other) using Lemon RDF model, then interlinked.



Work in Progress

- Lemon-izing and Interlinking Arabic resources with the Linguistic Linked Open Data Cloud
- Building an Arabic Knowledge Graph
- Collect and interlink dialect corpora

References

1. Mustafa Jarrar. The Arabic Ontology - An Arabic Wordnet with Ontologically Clean Content. Applied Ontology Journal, 2019 [Forthcoming].
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Mustafa Jarrar, Hamzeh Amayreh: An Arabic Multilingual Database with a Lexicographic Search Engine. The 24th International Conference on Applications of Natural Language to Information Systems (NLDB 2019), Springer. Manchester, UK. 2019
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4. Hamzeh Amayreh, Mohammad Dwaikat, and Mustafa Jarrar. Lexicon Digitization -A Framework for Structuring, Normalizing and Cleaning Lexical Entries, 2018.
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8. Mustafa Jarrar. Building a Formal Arabic Ontology (Invited Paper). In Proceedings of the Experts Meeting on Arabic Ontologies and Semantic Networks. ALECSO, Arab League, 2011